



DP-600 : Implement analytics solutions using Microsoft Fabric

Exercise 7 : Transform data with notebooks in Microsoft Fabric

- **Create a workspace**
- **Create a Lakehouse.**
- **Shape and clean the sales data**
- **Join and aggregate the data**
- **Apply window functions**
- **Write results to a Delta table**



DP-600 : Implement analytics solutions using Microsoft Fabric



Create a workspace

Create a workspace

Name *
Transform_data_notebook_7
This name is available

Description
Describe this workspace

Domain
Assign to a domain (optional)

[Learn more about workspace settings](#)

Workspace image
Upload
Reset

Advanced

Contact list *
rsidvika (Owner)

Workspace type
What a user can do in a workspace depends on the individual user license(s) assigned to that user and the workspace type. [Learn more](#)

Power BI only workspace types
Power BI workspaces only support the use of Power BI capabilities.

Apply Cancel

WS name: Transform_data_notebook_7

WS Type: Trial



DP-600 : Implement analytics solutions using Microsoft Fabric



Create a Lakehouse.

New Lakehouse

Name *

Location *

 Transform_data_notebook_7 ▼ ↗

☒ Lakehouse schemas ⓘ

CreateCancel

LH name: Sales_notebook



DP-600 : Implement analytics solutions using Microsoft Fabric

► Shape and clean the sales data



Shape and clean the sales data

Real-world data rarely arrives clean. The following query applies four transformations in a single pass:

- `SELECT DISTINCT` removes the duplicate row, reducing the count from 11 to 10.
- `COALESCE(region, 'Unknown')` replaces the null region with a default value.
- `quantity * unit_price AS line_total` adds a calculated column for the total sale amount.
- `CASE ... END AS value_tier` creates a conditional column that categorizes each order as High, Medium, or Low.



```
1  %%sql
2  -- Clean raw sales: remove duplicates, fill nulls, add calculated and conditional columns
3  CREATE OR REPLACE TEMP VIEW clean_sales AS
4  SELECT DISTINCT
5      order_id,
6      customer_id,
7      product_id,
8      quantity,
9      unit_price,
10     order_date,
11     COALESCE(region, 'Unknown') AS region,
12     ROUND(quantity * unit_price, 2) AS line_total,
13     CASE
14         WHEN quantity * unit_price > 100 THEN 'High'
15         WHEN quantity * unit_price > 50 THEN 'Medium'
16         ELSE 'Low'
17     END AS value_tier
18 FROM raw_sales
```

The screenshot shows the Microsoft Fabric interface. On the left is a sidebar with icons for Copilot, OneLake catalog, Monitor, Real-Time, and Workloads. The main area displays the query execution results. At the top, the query is shown with a status bar indicating it executed successfully in 18 seconds. Below the query, there's a section for Spark jobs. The main part of the interface shows a table view of the results, with columns for order_id, customer_id, product_id, quantity, unit_price, order_date, region, line_total, and value_tier. The table contains 10 rows of data.

	123 order_id	ABC customer_id	ABC product_id	123 quantity	12 unit_price	order_date	ABC region	12 line_total	ABC value_tier
1	1	C001	P001	5	29.99	2026-01-15	East	149.95	High
2	2	C002	P002	2	49.99	2026-01-20	West	99.98	Medium
3	3	C001	P003	1	199.99	2026-02-10	East	199.99	High
4	4	C003	P001	10	29.99	2026-02-14	North	299.9	High
5	5	C002	P004	3	15.99	2026-03-01	West	47.97	Low
6	6	C004	P002	1	49.99	2026-03-05	Unknown	49.99	Low
7	7	C001	P001	4	29.99	2026-03-15	East	119.96	High
8	8	C005	P003	2	199.99	2026-04-01	South	399.98	High
9	9	C003	P005	6	9.99	2026-04-10	North	59.94	Medium
10	10	C002	P001	3	29.99	2026-04-20	West	89.97	Medium



DP-600 : Implement analytics solutions using Microsoft Fabric



Join and aggregate the data

Home Edit AI tools Run View Comments History Develop Share

Run all Standard session PySpark (Python) Environment Workspace default Data Wrangler Copilot

Join and aggregate the data

Cleaned data becomes more useful when enriched with context from other tables. The next cell joins the sales data with customer and product reference tables using `INNER JOIN`.

```
1 %%sql
2 -- Join sales with customer and product tables
3 CREATE OR REPLACE TEMP VIEW enriched_sales AS
4 SELECT
5     s.order_id,
6     s.order_date,
7     c.customer_name,
8     p.product_name,
9     p.category,
10    s.quantity,
11    s.unit_price,
12    s.line_total,
13    s.value_tier,
14    s.region
15 FROM clean_sales s
16 INNER JOIN customers c ON s.customer_id = c.customer_id
17 INNER JOIN products p ON s.product_id = p.product_id
```

[4] ✓ 3 sec - Command executed in 3 sec 314 ms by [redacted] on 6/1/2026, 6:00:04 PM

Log

Copilot OneLake catalog Monitor Real-Time Workloads Transform_d ata_noteb...

Explorer

```
1 %%sql
2 -- Aggregate: regional revenue summary
3 SELECT
4     region,
5     COUNT(*) AS order_count,
6     ROUND(SUM(line_total), 2) AS total_revenue,
7     ROUND(AVG(line_total), 2) AS avg_order_value
8 FROM enriched_sales
9 GROUP BY region
10 ORDER BY total_revenue DESC
```

[5] ✓ 15 sec - Command executed in 15 sec 739 ms by sidvika minnu on 6/1/2026, 6:01:48 PM

> Spark jobs (14 of 14 succeeded) Resources Log

Table + New chart

Table view

	ABC region	12L order_count	12 total_revenue	12 avg_order_value
1	East	3	469.9	156.63
2	South	1	399.98	399.98
3	North	2	359.84	179.92
4	West	3	237.92	79.31
5	Unknown	1	49.99	49.99



DP-600 : Implement analytics solutions using Microsoft Fabric



Apply window functions

The screenshot shows the Microsoft Fabric notebook interface. The top navigation bar includes Home, Edit, AI tools, Run, and View. Below this is a toolbar with icons for saving, downloading, settings, running all cells, session status (Standard session), environment (PySpark (Python)), and workspace (Workspace default). The left sidebar contains the Explorer pane with icons for Home, Workspaces, Copilot, OneLake catalog, Monitor, Real-Time, and Workloads. The main area displays a notebook titled "Apply window functions" with a description: "Window functions calculate values across related rows without collapsing the detail. The next cell adds a running total OVER .". The notebook content shows a SQL query:

```
1 %%sql
2 -- Window functions: running total and order sequence per customer
3 SELECT
4     customer_name,
5     order_date,
6     line_total,
7     SUM(line_total) OVER (
8         PARTITION BY customer_name ORDER BY order_date
9     ) AS running_total,
10    ROW_NUMBER() OVER (
11        PARTITION BY customer_name ORDER BY order_date
12    ) AS order_sequence
13 FROM enriched_sales
14 ORDER BY customer_name, order_date
```

Below the query, a status bar indicates: [6] ✓ 7 sec - Command executed in 7 sec 485 ms by sidvika minnu on 6/1/2026, 6:12:45 PM. At the bottom, there are links for Spark jobs (5 of 5 succeeded), Resources, and Log.

Table view						
	ABC customer_name	order_date	12 line_total	12 running_total	123 order_sequence	
1	Adventure Works	2026-03-05	49.99	49.99	1	
2	Contoso Ltd	2026-01-15	149.95	149.95	1	
3	Contoso Ltd	2026-02-10	199.99	349.94	2	
4	Contoso Ltd	2026-03-15	119.96	469.9	3	
5	Fabrikam Inc	2026-01-20	99.98	99.98	1	
6	Fabrikam Inc	2026-03-01	47.97	147.95	2	
7	Fabrikam Inc	2026-04-20	89.97	237.92	3	
8	Northwind Traders	2026-02-14	299.9	299.9	1	
9	Northwind Traders	2026-04-10	59.94	359.84	2	
10	Woodgrove Bank	2026-04-01	399.98	399.98	1	



DP-600 : Implement analytics solutions using Microsoft Fabric



Write results to a Delta table

Write results to a Delta table

Persisting results as a Delta table makes the data available to reports, other notebooks, and downstream pipelines. The next

+ Code + Markdown

```
1 %%sql
2 -- Write enriched data to a permanent Delta table
3 CREATE OR REPLACE TABLE gold_sales AS
4 SELECT * FROM enriched_sales
```

[7] ✓ 9 sec - Command executed in 9 sec 255 ms by sidvika minnu on 6/1/2026, 6:16:25 PM

> Spark jobs (10 of 10 succeeded) Resources Log

Copilot

OneLake catalog

Monitor

Real-Time

Workloads

Transform data notebook...

...

Table

New chart

Table view

	ABC category	ABC region	12L orders	12 revenue
1	Electronics	South	1	399.98
2	Accessories	North	1	299.9
3	Accessories	East	2	269.91
4	Electronics	East	1	199.99
5	Accessories	West	2	137.94
6	Electronics	West	1	99.98
7	Components	North	1	59.94
8	Electronics	Unknown	1	49.99

1 %%sql

2 -- Verify the Delta table contents

3 SELECT category, region, COUNT(*) AS orders, ROUND(SUM(line_total), 2) AS revenue

4 FROM gold_sales

5 GROUP BY category, region

6 ORDER BY revenue DESC

[8] ✓ 3 sec - Command executed in 3 sec 342 ms by sidvika minnu on 6/1/2026, 6:16:36 PM

> Spark jobs (6 of 6 succeeded) Resources



DP-600 : Implement analytics solutions using Microsoft Fabric



Reference

